

C “ Master Traceability Matrix (browser extension)

Revision: 1 **Last modified:** 2026-06-10T00:00:00Z **Purpose:** Anti-skip backbone (operator mandate + §11.4.118 enumerated coverage). One row per atomic requirement / feature / entity / edge-case / NFR / dimension-decision drawn from ALL six analysis files (docs / browser_extension / _analysis / 01 . . 06). EVERY item from the source corpus appears. Proves nothing is silently dropped: deferred items are listed explicitly with rationale; untraceable items are listed OPEN.

Stable ID scheme: - FR-001 . . FR-025 “ 25 functional requirements (file 01 §B). - NFR-001 . . NFR-015 “ 15 non-functional requirements (file 01 §C). - DIM01-Rxx . . DIM12-Rxx “ per-dimension consolidated decisions (file 02). - ER- <entity> “ 6 ER entities (file 03 §2) + SQL-table superset (file 04 §4a). - EDGE-xx “ documented edge cases (file 01 §I + scattered). - SEC-xx “ security/STRIDE controls (file 02 Dim11 + file 03 §6 + file 01 §N). - INS-xx “ 10 cross-dimension insights (file 02). - CV-xx “ cross-verification conflicts/low-confidence items (file 02). - FND-xx “ foundation/source-layer facts (files 04/05). - TST-xx “ test-type coverage requirements (file 06). - BLD-xx “ build/release/distribution (file 01 §M + file 02 Dim12 + file 03 §7). - OQ-xx “ open questions / ambiguities (all files).

Phase column uses the authoritative 8-phase impl-plan (file 01 §P.2): P1 Foundation, P2 Core Engine, P3 Extension Shell, P4 API Integration, P5 Tab Groups, P6 UI/UX Polish, P7 Testing, P8 Build/Dist. v2 = deferred (see Coverage Ledger). Boba constraint flags are noted in Notes.

1. Functional Requirements (25)

ID	Source Requirement	Category	Phase	Notes/Dependencies
FR-001	01 Detect magnet links (href magnet:?) in DOM of any page, §B.1 dynamic across SPA; iframe via	detection	P2/P3	MutationObserver debounce 500 ms (EDGE-06); regex MAGNET_REGEX (FND-12)
	FR-001 all_frames; %500 ms after injection			
FR-002	01 Detect .torrent file links via §B.1 .torrent suffix (case-insens) +	detection	P2/P3	TORRENT_FILE_REGEX (FND-12); CORS-aware (DIM07)
	FR-002 Content-Type: application/x-bittorrent			
FR-003	01 Parse magnet URI per BEP-9: xt/ §B.1 dn/tr/xl/ws/x.pe; xt mandatory valid	parsing	P2	OQ-07 param-set drift (xs/as/kt/mt vs x.pe); DIM06
	FR-003 Torrentâ€¢; malformed tr discarded			
FR-004	01 Download + parse .torrent via §B.1 fetch; max 10 MB	parsing/ api	P2/P4	EDGE-04, EDGE-05; FND parser is dead code in ref (FND-23)
	FR-004 (maxTorrentFileSize); 30 s timeout; CORSâ†’Boba-proxy fallback; bencode decode			
FR-005		parsing	P2	

ID	Source Requirement	Category	Phase	Notes/Dependencies
	01 Compute SHA-1 over bencoded Â§B.1 info dict â†’ 40-hex infohash; FR-005 MUST match xt			RAW bencode, not decode- re-encode (DIM07-R; FND-22)
FR-006	01 Deduplicate torrents by lowercase Â§B.1 infohash; merge longest FR-006 displayName + union trackers + earliest detectedAt	parsing	P2	INS-02 unified identity; ref time-salted ids unreliable (OQ-16)
FR-007	01 Send magnets to qBitTorrent / Â§B.2 api/v2/torrents/add_urls FR-007 param; cookie session; opt category/tags/savepath/rename	api	P4	DIM02; Bobaâ†’7186 proxy (OQ-01)
FR-008	01 Upload .torrent via multipart Â§B.2 torrents field to /api/v2/ FR-008 torrents/add; blobs in- memory only	api	P4	addTorrentFromFile (FND-31)
FR-009	01 Tab-group batch: enumerate group Â§B.2 tabs, SCAN each, aggregate + FR-009 dedup, transmit unified batch	tab- groups	P5	NET-NEW design â€” ref has NO tabGroups code (OQ-17/ FND-33)
FR-010	01 Enumerate tab groups + extract Â§B.2 URLs via FR-010 chrome.tabGroups.query() + chrome.tabs.query(); show titles/colors; multi-select	tab- groups	P5	DIM05; Chrome/Yandex only (EDGE-21)
FR-011	01 Auto-discover Boba on LAN: Â§B.3 probe well-known ports + / FR-011 health; parse version â‰¥1.0.0; ranked list; persist	api	P4	Re-point 8443/8080â†’7186/7187/7189 (OQ-01); INS-01
FR-012	01 Support 4 auth methods: cookie Â§B.3 (SID), API key (X-API-Key), FR-012 Basic, Custom Header	security/ api	P4	C1: support both cookie+JWT/apikey; ref drops custom-header (OQ-06)
FR-013	01 Encrypt stored credentials Â§B.3 AES-256-GCM; PBKDF2 from FR-013 per-install salt; key in storage.session; 96-bit IV; 128-bit tag	security	P1/P4	Â§11.4.10; ref uses fixed/ empty passphrase BLUFF (OQ-15/FND-30)
FR-014	01 Offline FIFO queue: max 1000; exp Â§B.4 backoff 5 sâ†’5 min; max 5 retries; FR-014 persist storage.local; Dead-Letter substore	queue	P4	ref max 50 / 3 attempts diverges (OQ-19); DIM09
FR-015	01 Badge: numeric queued count; Â§B.4 colors green idle/blue sending/ FR-015 orange queued/red error; â‰¥200 ms update	ui	P3/P6	INS-09 badge sync; BADGE_COLORS (FND-26)
FR-016	01 Context menu: Â§B.5 link(magnet)â†’Send Magnet; FR-016 link(.torrent)â†’Download; pageâ†’Scan; tab-groupâ†’Send Group	ui	P3/P5	chrome.contextMenus; ref bobalink-send/ scan/dashboard (FND-32)
FR-017	Keyboard shortcuts: send-current- page Ctrl+Shift+B; open-popup	ui	P3	OQ-08 3-vs-4 cmds; ref ships B/S/D not L/G (FND-29)

ID	Source	Requirement	Category	Phase	Notes/Dependencies
	01 Â§B.5 FR-017	Ctrl+Shift+L; send-tab-group Ctrl+Shift+G (customizable)			
FR-018	01 Â§B.5 FR-018	Cross-browser single codebase: Chrome 88+, Firefox 109+, Opera 74+, Yandex 21+; BrowserAdapter abstraction	cross- browser	P1/P8	DIM04; WXT handles per- browser manifest
FR-019	01 Â§B.5 FR-019	il8n via <code>_locales/{lang}/ messages.json</code> : en default + zh_CN/zh_TW/es/fr/de/ja/ko; no hardcoded strings	il8n	P6	ref ships en only (28 keys, FND-19)
FR-020	01 Â§B.5 FR-020	Accessibility WCAG 2.1 AA: contrast 4.5:1; keyboard nav; ARIA on icon buttons; aria-live; focus indicators	ui	P6	Â§11.4.117 pixel-oracle for verification
FR-021	01 Â§B.5 FR-021	Dark/Light theme: respect <code>prefers-color-scheme</code> + manual override; CSS custom properties	ui	P6	DIM10 Â§9; ref <code>ui.theme=auto</code> (FND-25)
FR-022	01 Â§B.5 FR-022	Notifications via <code>chrome.notifications</code> : Sent / Send Failed(retry) / Batch Complete / Retrying; suppressible per-category	ui	P3/P6	DIM10 Â§5; MV3 no base64 icons (DIM03)
FR-023	01 Â§B.5 FR-023	Options page: server URL+mode, <code>auth(masked)+test-connection</code> , default category/tags/path, notif prefs, queue settings, theme, reset	ui	P3	DIM10 Â§2; ref Servers/ General/Advanced/About (FND-43)
FR-024	01 Â§B.5 FR-024	Health check: poll 30 s; <code>/api/ v2/app/version</code> or <code>/health</code> ; states connected/connecting/ disconnected/error; tooltip last-ping	api	P4	ref <code>HealthChecker</code> thresholds 2 s/5 s (FND-34)
FR-025	01 Â§B.5 FR-025	Rate limiting: 10 req/s burst, 60/ min sustained; 1 concurrent + 500 ms inter-call; honor 429 Retry- After; in-memory	api	P4	Token-bucket (file 01 Â§S); ref <code>RATE_LIMIT</code> 10/1000ms (FND-27)

2. Non-Functional Requirements (15)

ID	Source	Requirement	Category	Phase	Notes/ Dependencies
NFR-001	01 Â§C.1	Content script init 10 ms (<code>performance.now()</code>)	performance	P7	benchmark test (TST-09)
NFR-002	01 Â§C.1	Magnet detection 5 ms/link (1,000-link page benchmark)	performance	P7	DOM-first querySelector (INS-03)
NFR-003	01 Â§C.1	Popup render 100 ms (Lighthouse)	performance	P7	

ID	Source	Requirement	Category	Phase	Notes/ Dependencies
NFR-004	01 Â§C.1	Options page load â€œ200 ms (Lighthouse)	performance	P7	
NFR-005	01 Â§C.1	API round-trip â€œ500 ms local (p95 over 100 calls)	performance	P7	
NFR-006	01 Â§C.1	Service worker cold start â€œ50 ms	performance	P7	
NFR-007	01 Â§C.1	Bundle size compressed â€œ350 KB	build	P8	tree-shake + code-split
NFR-008	01 Â§C.2	Credential encryption at rest: AES-256-GCM via Web Crypto	security	P1/P4	composes FR-013/ SEC-04
NFR-009	01 Â§C.2	No cleartext credential logging (ESLint rule + review)	security	P1	Â§11.4.10; logger must redact
NFR-010	01 Â§C.2	HTTPS-only API comms (URL scheme validation + CSP)	security	P4	except localhost (SEC-08)
NFR-011	01 Â§C.2	CSP script-src 'self'; object-src 'none'	security	P1	MV3 CSP object (SEC-06)
NFR-012	01 Â§C.2	Minimum permission model: activeTab + declared host perms only	security	P1	OQ-09 activeTab vs https://*/ ; ref over-broad (FND-08)
NFR-013	01 Â§C.3	Crash-free session rate â€œ99.9%	reliability	P7	chaos/stress (TST-07/08)
NFR-014	01 Â§C.3	Offline queue durability 100% â€œall items survive restart	reliability	P4/P7	persist storage.local; chaos test
NFR-015	01 Â§C.3	API compatibility qBitTorrent 4.4.xâ€œ5.x	api	P4/P7	version matrix (file 01 Â§R); v5 state renames (DIM02)

3. Research-Dimension Consolidated Decisions (12 dimensions)

ID	Source Requirement / Decision	Category	Phase	Notes/ Dependencies
DIM01- 02 R1	Treat Dim01 as authoritative Boba topology: qBt 7185 (proxy 7186), FastAPI 7187, bridge 7188, jackett 9117, boba-jackett 7189; Go proxy 7187 (profile go)	data-model/ api	P1/P4	overrides spec 8443/8080 (OQ-01)
DIM01- 02 R2	Extension talks to FastAPI :7187: /api/v1/search(+SSE stream), /api/v1/download, /api/v1/download/file, /api/v1/magnet, /api/v1/auth/status, /api/v1/config, /health	api	P4	confirm vs live OpenAPI (OQ-12)
DIM01- 02 R3	Boba CORS via ALLOWED_ORIGINS env incl. chrome-extension://<id>; SW not subject to CORS	security/api	P4	INS-05 satellite
DIM01- 02 R4	qBt default creds admin/admin (cookie SID); private trackers env-driven; 48 nova3 plugins; freeleech-only IPTorrents	api	P4	K.3 never commit secrets
DIM02- 02 R1	qBt WebUI v2: cookie auth /api/v2/auth/loginâ†’SID; Referer/Origin MUST match Host (CSRF); 403=IP ban	api/security	P4	exp backoff to avoid bans
DIM02- 02 R2	/torrents/add: magnets in urls, .torrent multipart torrents; 415 invalid; 200â‰‰ guaranteed-added; 405 wrong-method v4.4.4+	api	P4	
DIM02- 02 R3	API-key auth (Bearer qbt_) v5.2.0+ only; cookie is safe baseline	api	P4	CV low-confidence (CV-03)
DIM02- 02 R4	Monitoring via polling /sync/maindata?rid= (no native WS/SSE); v5 renames pausedUPâ†’stoppedUP/pausedDLâ†’stoppedDL MV3 manifest: host_permissions separate; background	api	P4	INS-09 badge sync
DIM03- 02 R1	{service_worker, type: module}; perms storage/notifications/contextMenus/scripting/alarms/activeTab	cross-browser	P1	
DIM03- 02 R2	SW HARD timeouts: 30 s idle / 5 min request / 30 s fetch; chrome.alarms keep-alive; register listeners synchronously top-level; return true for async	cross-browser	P3	EDGE-15; ref keepalive 20 s (FND)
DIM03- 02 R3	Content scripts document_idle; ISOLATED world; cross-origin fetch routed through SW (CS canâ€™t since Chrome 85)	detection/ security	P3	DIM07-R4
DIM03- 02 R4	Storage quotas: local ~10 MB, sync ~100 KB/8 KB-item, session ~10 MB in-memory; badge â‰‰4 chars	data-model	P1	
DIM03- 02 R5	Offscreen documents for DOM/parse in MV3 (one/ profile); NOT in Firefox	parsing	P2	EDGE Firefox fallback
			P1/P8	C2/C3 resolved

ID	Source	Requirement / Decision	Category	Phase	Notes/ Dependencies
DIM04-R1	02 Dim04	MV3 everywhere; Firefox uses Event Pages (WXT handles); webextension-polyfill for browser.* on Chromium + own feature-detect	cross-browser		
DIM04-R2	02 Dim04	Firefox MV3 REQUIRES gecko.id + strict_min_version; version x.y.z only; data_collection_permissions since Nov 2025	build	P8	BLD
DIM04-R3	02 Dim04	Opera lacks storage.sync/storage.managed; notifications partial on Mac; sidebarAction Opera/FF only	cross-browser	P6/P8	compat matrix (ER/diagram #12)
DIM04-R4	02 Dim04	UA detection order Yandex†'Opera†'Edge†'Chrome†'Chromium; prefer feature detection	cross-browser	P1	
DIM05-R1	02 Dim05	tabGroups Chrome 89+/MV3; query/get/update/move; create via tabs.group/ungroup; groupId NOT persistent (id by title+color+URLs)	tab-groups	P5	EDGE-21; Firefox incompatible
DIM05-R2	02 Dim05	Accessing tab.url/title needs tabs perm or host perms; collapsed groups still API-accessible	tab-groups	P5	conflicts NFR-012 minimal perms
DIM05-R3	02 Dim05	Yandex Chromium ~147+ full tabGroups; Chrome 148+ chrome://newtab cannot be grouped	tab-groups	P5	EDGE-21
DIM06-R1	02 Dim06	Magnet BEP-9: xt only mandatory; hash v1 40-hex / 32-base32 (convert) / v2 btmh 1220+64-hex; two-phase detect+validate	parsing	P2	base32†'hex (DIM06-R2)
DIM06-R2	02 Dim06	base32 (32-char) infohash MUST convert to hex (40-char); RFC4648 alphabet	parsing	P2	base32ToHex (FND)
DIM06-R3	02 Dim06	Decode dn (decodeURIComponent + +†'space); tr/xs/as/ws decode; kt split +; so BEP-53 ranges; duplicate params†'array	parsing	P2	
DIM06-R4	02 Dim06	Sanitize dn before HTML render (strip control/<>, â%□255, textContent); chunk large text 100 KB vs backtracking	security/ parsing	P2	SEC-10
DIM07-R1	02 Dim07	.torrent = bencode dict announce+info; info required piece length/pieces(20-byte SHA-1)/name; single length XOR multi files[]	parsing	P2	
DIM07-R2	02 Dim07	Bencode: dict keys lexicographic RAW-byte sorted (load-bearing for infohash); strict type rules	parsing	P2	TST bencode
DIM07-R3	02 Dim07	Infohash = SHA-1 of RAW bencoded info dict as-in-file (never decode-re-encode); Web Crypto needs HTTPS secure ctx	parsing	P2	FND-22 ref re-encodes (caveat)
DIM07-R4	02 Dim07	CORS download: CS canâ€™t cross-origin fetch since Chrome 85 â†' delegate to SW; size â%□10 MB (50 MB max); validate first byte 0x64	api/parsing	P2/P4	EDGE-05
DIM07-R5	02 Dim07	Private torrents (BEP-27): info.private==1 disables DHT/PEX; sanitize passkey to PASSKEY (never log)	security	P2	SEC; Â§11.4.10
DIM08-R1	02 Dim08	Content-script timing document_idle + readyState checks for heavy SPAs (safeInitialize)	detection	P3	
DIM08-R2	02 Dim08	DOM-first querySelectorAll(â€~a[href^=â€œmagnet:â€¢]â€™) detection	detection	P2	INS-03 tier-1+tier-3

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	(~90% catch) + TreeWalker text-node scan skipping script/style			
DIM08-02 R3	Dim08 MutationObserver debounced 500 ms (~88Å— faster than polling), incremental scan new nodes; IntersectionObserver infinite scroll	detection	P2	EDGE-06
DIM08-02 R4	Dim08 iframe all_frames:true + match_about_blank:true; Shadow DOM recursive shadowRoot (querySelectorAll doesnâ€™™t pierce)	detection	P2	EDGE-07
DIM08-02 R5	Dim08 Site-specific selector DB top ~20 sites (TPB/1337x/nyaa/rutracker/generic) â€” tier-2 detection	detection	P2	INS-03; FND two tables to reconcile (OQ)
DIM08-02 R6	Dim08 Scanner cleanup: track observers/timers/listeners, disconnect on navigation (memory-leak prevention); 16 ms frame budget	performance	P2	NFR-SCAL-3
DIM09-02 R1	Dim09 Communication patterns: fetch (extâ†’server), EventSource SSE, WebSocket progress, runtime.sendMessage, storage.local, alarms	api	P4	
DIM09-02 R2	Dim09 Auth strategies: cookie/SID, JWT Bearer (60 s buffer + single-flight refresh, password NOT stored), API key, Basic	security/api	P4	C1 support both
DIM09-02 R3	Dim09 Retry classification: retry 0/429/500/502/503/504/401-after-refresh; non-retry 400/403/404/415; exp backoff+jitter (base 1000, max 30000, Å—2, jitter 0.5, max 3)	queue/api	P4	file 01 Å§F.3 differs (5 retries) (OQ-19)
DIM09-02 R4	Dim09 SSE/EventSource cannot send custom headers â€” auth via URL query param; reconnect exp backoff; Chrome 116+ keeps SW alive on WebSocket	api	P4	CV-low SSE path (CV-04)
DIM09-02 R5	Dim09 Credential storage: store tokens only never plaintext password; chrome.storage.local NOT encrypted by default; clear on logout; HTTPS+validate certs	security	P4	FR-013 adds encryption
DIM09-02 R6	Dim09 Health: separate liveness/readiness; parallel checks AbortSignal.timeout(5000); status healthy/degraded/unhealthy	api	P4	
DIM09-02 R7	Dim09 Offline queue persist + retry on reconnect; navigator.onLine unreliable for LAN â†’ proactive / app/ver s i o n ping every 30 s via alarms	queue	P4	INS-06
DIM09-02 R8	Dim09 Rate limiting token-bucket per-endpoint; RequestBatcher (maxBatch 10, maxWait 50 ms)	api	P4	FR-025
DIM09-02 R9	Dim09 Manifest host_permissions localhost+127.0.0.1 (http+https * port); externally_connectable localhost; NEVER <a l l _ u r l s > in prod	security	P1	NFR-012; OQ-09
DIM10-02 R1	Dim10 Popup Chrome max 800Å—600 / min 25Å—25; sweet spot 380â€“450px wide; fixed header/connection bar/scroll/footer	ui	P3	FND popup 400px
DIM10-02 R2	Dim10 Context menus require contextMenus perm; >1 visible item auto-collapses into extension-named parent	ui	P3	
		ui	P3	FR-015

ID	Source Requirement / Decision	Category	Phase	Notes/ Dependencies
DIM10-02 R3	Badge text â‰¥4 chars; per-tab via tabId (resets on close); setBadgeText/Color			
DIM10-02 R4	Commands max 4 suggested keys; all need Ctrl/Alt; Mac Ctrlâ†‘Command default; fires in SW	ui	P3	OQ-08
DIM10-02 R5	Side panel Chrome 114+ (open 116+, close 141+); cannot auto-open without user gesture	ui	v2	CV-low (CV-05); optional surface
DIM10-02 R6	i18n_locales/<locale>/messages.json; manifest default_locale required if _locales; __MSG_key__+getMessage(); @@bidi for RTL	i18n	P6	FR-019
DIM11-02 R1	Design principles: Least Privilege, Defense in Depth, Zero Trust, Secure by Default, Privacy First	security	P1	SEC umbrella
DIM11-02 R2	STRIDE mitigations S1-4/T1-4/R1-2/I1-5/D1-3/E1-3 + trust boundaries TB1-4	security	P1/P7	SEC-xx; security test (TST-04)
DIM11-02 R3	Minimal perms storage/alarms/notifications/activeTab; optional clipboardWrite; host localhost; NEVER all_urls/tabs/cookies/webRequest/scripting/downloads/history	security	P1	NFR-012; ref declares scripting (FND-08)
DIM11-02 R4	Specific per-site matches (never wildcard â€” wildcards nullify postMessage protection)	security	P1	conflicts ref <all_urls> CS (FND-08)
DIM11-02 R5	AES-256-GCM + PBKDF2-HMAC-SHA256 100k iters; 128-bit salt; 96-bit IV; non-extractable key; master pw never stored; decrypted cache storage.session; Clear All Data (GDPR)	security	P4	FR-013/NFR-008
DIM11-02 R6	Input validation: magnet/info-hash 40-hex v1/64-hex v2; sanitize dn textContent; validate Boba URL; reject javascript:/data:/vbscript;; block dangerous ports	security	P2/P4	SEC-10
DIM11-02 R7	Privacy: collect only Boba URL/creds(encrypted) + ephemeral magnet/names; NO page content/history/IP; analytics opt-in; eraseAllData(); no 3rd-party/selling	privacy	P6	defaults analytics false
DIM11-02 R8	Secure fetch: credentials: 'omit' + referrerPolicy: 'no-referrer'; HTTPS-only except localhost; HSTS; self-signed cert pinning/CA install	security	P4	conflicts qBt needs credentials:include (FND-31) â€” reconcile
DIM11-02 R9	Validate sender.id + sender.url (chrome-extension://) on ALL onMessage; action allow-listing	security	P3	ref does NOT validate sender (OQ)
DIM11-02 R10	Update security: official-store signed distribution; FF self-hosted update_hash HTTPS; verify perms unchanged on update	build	P8	
DIM12-02 R1	Build WXT (Vite); test Jest unit + Playwright E2E; mock sinon-chrome; versioning Conventional Commitsâ†‘SemVer	build/testing	P7/P8	OQ Jest-vs-Vitest (OQ-03)
DIM12-02 R2		testing	P7	E2E fixture (TST-02)

ID	Source Requirement / Decision	Category	Phase	Notes/Dependencies
	Playwright needs launchPersistentContext + “load-extension + “disable-extensions-except; jsdom for DOM tests			
DIM12-02 R3	Dim12 Coverage threshold global 70% config (aspirational 80%+ INS-10); reporters lcov/html/json-summary; optional Sonar/Snyk/Semgrep/Trivy/Gitleaks	testing	P7	Boba’s ~100% types
DIM12-02 R4	Dim12 Store assets: icons 16/32/48/128(+512); screenshots 640—480 or 1280—800; privacy policy; no console.log/source-maps in prod; CSP set; no remote code	build	P8	BLD

4. ER Entities / Data Model (6 ER + SQL superset)

ID	Source	Entity / Requirement	Category	Phase	Notes/Dependencies
ER-extension_config	03 §2; 04 §4a T2	Single per-install settings (config_id PK, server_url, api_key, qbittorrent_url, username, password_encrypted, auto_detect, default_category, scan_interval_ms, max_retries, retry_delay_ms, timestamps)	data-model	P1	SQL key-value form (25 seed rows); OQ FK to discovered (diagram #9)
ER-discovered_torrents	03 §2; 04 §4a T3	Torrents found on pages (id=infohash PK UK, magnet_uri, name, size_bytes, source_url, source_tab_id, status, discovered_at, sent_at, error_message; SQL adds discovery_method/ is_private/tab_group_id/ selected/boba_status)	data-model	P2/P3	mmd authoritative superset (file 03 note)
ER-download_queue	03 §2; 04 §4a T4	Offline/retry queue (queue_id PK, torrent_id FK, magnet_uri, priority 0-10, retry_count, status queued/retrying/failed/ completed, queued_at, last_retry_at, error_log; SQL adds next_retry_at/ server_id/torrent_data_b64)	queue/ data-model	P4	FR-014
ER-server_config	03 §2; 04 §4a T5	Per-Boba-server multi-server (server_id PK, name, fastapi_url, go_service_url, qbittorrent_url, api_key, is_active, health_check_interval,	data-model	P1/P4	DIM01-R1 port reconcile

ID	Source	Entity / Requirement	Category	Phase	Notes/Dependencies
ER-send_history	03 Â§2; 04 Â§4a T6	last_health_check, last_check_result; SQL ports 7187/7189/8080â†’7186) Audit log every send (history_id PK, infohash, magnet_uri, name, server_id FK, success, qbittorrent_hash, category, tags, sent_at, response_time_ms, error_message)	data- model	P4	immutable log
ER-search_cache	03 Â§2; 04 Â§4a (n/a)	Cached search results TTLâ€™™d (cache_id PK, query_hash, query_text, results JSON, result_count, cached_at, ttl_seconds, is_valid) â€™ standalone, no relationship	data- model	v2	search aggregation gated Boba 1.2+; OQ-12 storage backend
ER-REL	03 Â§2 relationships	Cardinality: extension_config 1â€™”o{ discovered_torrents; discovered_torrents 1â€™”o{ download_queue; server_config 1â€™”o{ send_history	data- model	P1	extension_configâ†’discovered FK conceptual only (OQ)
ER-SQL-extra	04 Â§4a T1/ T7/T8/T9	SQL superset tables: app_metadata, site_selectors (30 seed rows), queue_log, statistics (daily aggregates)	data- model	P1/P2	sql.js-over-storage.local OR chrome.storage.JSON â€™ pick one (OQ dual-storage)

5. Edge Cases (documented)

ID	Source	Edge case	Category	Phase	Notes
EDGE-01	01 Â§I	Magnet without dn â†’ â€™œUnknown Torrentâ€™œ; qBt resolves from metadata	parsing	P2	FR-003
EDGE-02	01 Â§I	Duplicate infohash â†’ silently skipped â€™œduplicateâ€™œ; merge longest dn+union tr+earliest ts	parsing	P2	FR-006
EDGE-03	01 Â§I	Malformed tracker URLs â†’ logged + discarded	parsing	P2	FR-003
EDGE-04	01 Â§I	.torrent >10 MB â†’ not downloaded; E_FILE_TOO_LARGE	api	P4	FR-004

ID	Source	Edge case	Category	Phase	Notes
EDGE-05	01 Â§I	413 â†’ â€œuse magnetâ€” CORS-blocked .torrent â†’ Boba proxies (Boba mode); qBt-direct manual copy	api	P4	FR-004; E_CORS
EDGE-06	01 Â§I	Dynamic SPA/infinite-scroll links â†’ MutationObserver debounce 250/500 ms	detection	P2	DIM08-R3
EDGE-07	01 Â§I	iframe links â†’ only when all_frames; may be inaccessible	detection	P2	DIM08-R4
EDGE-08	01 Â§I	JS click-handler â€œlinksâ€” (not real <a>) â†’ not detected; manual copy	detection	P3	documented limitation
EDGE-09	01 Â§I	Non-standard/encoded magnet formats â†’ not detected; manual copy	detection	P2	
EDGE-10	01 Â§I	Login-gated / logged-in-only sites	detection	P3	documented
EDGE-11	01 Â§I	Strict-CSP pages â†’ extension respects CSP, will NOT inject scripts	detection/security	P3	threat model
EDGE-12	01 Â§I	Forgotten encryption password â†’ no recovery; reset extension + reconfigure	security	P4	FR-013
EDGE-13	01 Â§I	Pages 10,000+ links â†’ no jank (>16 ms); requestIdleCallback >500; pagination	performance	P2	NFR-SCAL-3
EDGE-14	01 Â§I	Browser restart w/o password â†’ session key lost â†’ re-enter credentials	security	P4	storage.session loss
EDGE-15	01 Â§I	Storage quota exceeded â†’ E_STORAGE_FULL; Firefox stricter; clear old queue	data-model	P4	DIM03-R4
EDGE-16	01 Â§I	MV3 SW termination â†’ chrome.alarms keep-alive; Firefox suspends more aggressively	cross-browser	P3	DIM03-R2
EDGE-17	01 Â§I	qBt rejects duplicate â†’ already	api	P4	FR-007

ID	Source	Edge case	Category	Phase	Notes
EDGE-18	01 Â§I	downloading/ completed Private-tracker torrents â†’ work if accessible; may need passkey in tracker URL	api/security	P4	DIM07-R5
EDGE-19	01 Â§I	Self-signed certs â†’ browser exception / system trust; HTTPS- only temporarily disabled for testing	security	P4	DIM11-R8
EDGE-20	01 Â§I	macOS __MACOSX metadata folder in ZIP â†’ load error, must delete	build	P8	
EDGE-21	01 Â§I	Mobile browsers unsupported; Safari only Partial MV3 not tested; Firefox tab- groups incompatible	cross-browser	P8	DIM05-R1
EDGE-22	02 Dim06/08	URL-encoded magnet hrefs; javascript: URLs containing magnets; data-* attrs; base32 hashes	detection	P2	DIM06/08
EDGE-23	02 Dim07	Empty/invalid bencode, missing info, leading zeros, neg-zero, out-of- order keys, unicode/ binary filenames, unterminated structures	parsing	P2	TST bencode
EDGE-24	02 Dim05	groupId reuse across sessions; incognito groups donâ€™t persist; pinned tabs auto-unpinned; max ~32-64 groups/window	tab-groups	P5	DIM05

6. Security / STRIDE Controls

ID	Source	Control	Category	Phase	Notes
SEC-01	01 Â§N; 02 Dim11	Threat model: credential theft, MITM, XSS, privilege escalation, fingerprinting	security	P1	DIM11-R2
SEC-02	02 Dim11 S	Spoofing: cert pinning/ HTTPS, origin validation, isolated world, magnet hash validation	security	P1/P4	
SEC-03	02 Dim11 T		security	P1/P8	

ID	Source	Control	Category	Phase	Notes
		Tampering: store-signed packages, HTTPS/HSTS, AES-GCM, infohash validation			
SEC-04	01 Â§N; 02 Dim11 I	Info disclosure: AES-256-GCM creds; minimal matches no all_urls; restrict web_accessible_resources + use_dynamic_url; sender validation; no telemetry	security	P1/P4	FR-013
SEC-05	02 Dim11 D/E	DoS: client rate-limit, document_idle, storage quotas; EoP: sender validation + action allow-list, no eval, strict CSP	security	P1/P4	
SEC-06	01 Â§N; 02 Dim11	CSP object: extension_pages script-src 'self'; object-src 'self'; connect-src 'self' https:	security	P1	NFR-011; ref CSP (FND-09)
SEC-07	03 Â§6	Crypto ops in Offscreen Document (Web Crypto SubtleCrypto.encrypt) + PBKDF2 key derivation	security	P4	DIM03-R5; Firefox bg-page fallback
SEC-08	02 Dim11; 03 Â§6	Network layer HTTPS-only TLS 1.2+ (except localhost); cert validation no self-signed / pinning	security	P4	DIM11-R8
SEC-09	03 Â§6	Boba-services controls: API-key auth (X-API-Key), Basic auth, rate-limit 100 req/min/key	security	P4	diagram #10
SEC-10	02 Dim06/ Dim11	Sanitize display names textContent; reject javascript:/data:/vbscript;; block dangerous ports; chunk text vs backtracking	security	P2	DIM11-R6
SEC-11	02 Dim11	Privacy policy + opt-in analytics/error-reporting; no third-party sharing/ selling; Limited-Use compliance	privacy	P6/P8	DIM11-R7

7. Cross-Dimension Insights (10)

ID	Source	Insight	Category	Phase	Notes
INS-01	02 Insight 1	Zero-config discovery via well-known localhost ports (7187/7189/8080â†’7186) + version pings â†’ one-click Auto-Discover	api	P4	FR-011
INS-02	02 Insight 2	Unified torrent identity {infoHash, name, source, type}; magnet + .torrent converge on 40-char BTIH; dedup across tab group	parsing	P2	FR-006
INS-03	02 Insight 3	3-tier progressive detection: universal link scan (~90%) + site-specific selectors + text-based + MutationObserver	detection	P2	DIM08
INS-04	02 Insight 4	Tab group as batch job: right-click group â†’ extract all URLs â†’ parse â†’ submit batch newline-separated	tab-groups	P5	FR-009
INS-05	02 Insight 5	Extension as Boba satellite (â€œsend-toâ€œ), not management tool; use Boba auth/config; extend not duplicate UI (~30% less code)	api	P1/P4	architecture decision
INS-06	02 Insight 6	Offline-aware queue; navigator.onLine unreliable for LAN â†’ proactive / app / version ping 30 s via alarms + persistent queue	queue	P4	FR-014; DIM09-R7
INS-07	02 Insight 7	Privacy-first activeTab + optional host perms + scripting.executeScript on demand (MEDIUM â€” reviewers may prefer declared host perms)	security	P1	OQ-09 decision
INS-08	02 Insight 8	Multi-client gateway: TorrentClient interface (qBt/Transmission/Deluge/rTorrent)	data-model v2		DEFERRED â€” adds complexity; v1 leaves seam, targets Boba only
INS-09	02 Insight 9	Real-time badge sync: poll / sync / ma i n d a t a incremental RID â†’ badge count/color	ui	P3/P4	FR-015
INS-10	02 Insight 10	Enterprise-grade testing: 80%+ coverage, real-Boba-in-Docker E2E, security scanners, signed releases	testing	P7	TST-15; Boba mandate

8. Cross-Verification Conflicts & Low-Confidence Items

ID	Source	Item	Category	Phase	Notes
CV-01	02 cross-verif C1	Auth for Boba APIs: cookie (qBt direct) AND JWT/	security/api	P4	RESOLVED both valid; FR-012

ID	Source	Item	Category	Phase	Notes
CV-02	02 cross-verif C2/C3	API-key (Boba own) â€” support BOTH Build tool WXT (built on Vite); MV3 everywhere (WXT handles FF Event Pages)	build/cross-browser	P1	RESOLVED
CV-03	02 cross-verif low	qBt API-key Bearer only v5.2.0+; cookie baseline	api	P4	low-confidence; DIM02-R3
CV-04	02 cross-verif low/C4	Boba SSE path /api/v1/search/stream/{id} not confirmed in official docs; ports user-configurable	api	P4	confirm vs live OpenAPI (OQ-12)
CV-05	02 cross-verif low	Firefox E2E Playwright extension-ID extraction limits; side panel Chrome 114+ limits	testing/ui	P7/v2	TST-03; DIM10-R5

9. Foundation / Source-Layer Facts (load-bearing for the port)

ID	Source	Fact / Requirement	Category	Phase	Notes
FND-01	04 Â§1	Reference name bobalink v1.0.0 MIT (impl-plan says Apache-2.0 â€” OQ-10); rename for Boba (OQ naming)	build	P1	LICENSE + name decision
FND-02	04 Â§2	host_permissions 7187/7189/8080 â€” must change 8080â†’7186 (real qBt proxy)	cross-browser/ api	P1	OQ-01 critical
FND-03	04 Â§3	TS strict posture: all strict flags + noUncheckedIndexedAccess + exactOptionalPropertyTypes; zero any (no-explicit-any error)	build	P1	NFR-MAINT
FND-04	04 Â§3	ESLint type-aware: no-floating-promises/ no-misused-promises error; no-console warn (allow error/warn/info)	build	P1	NFR-009
FND-05			testing	P7	

ID	Source	Fact / Requirement	Category	Phase	Notes
	04 Â§3	Jest preset ts-jest ESM jsdom; coverage 80%; collectCoverageFrom EXCLUDES popup/options/content/background/index.ts			TST-15 must include UI
FND-06	04 Â§3	Playwright baseUrl chrome-extension://test-id (placeholder); â€œload-extension=../dist; chromium+firefox projects	testing	P7	TST-02; global-setup absent
FND-07	04 Â§2	content_scripts matches <all_urls> run_at document_idle (ref) â€” reconcile with minimal-perms	detection/ security	P1/P3	DIM11-R4 conflict
FND-08	04 Â§2; 05 Â§9	ref CSP connect-src localhost 7187/7189/8080; permissions include scripting (DIM11 says avoid)	security	P1	OQ-09
FND-09	04 Â§6	Regex set: MAGNET_REGEX/ MAGNET_VALIDATION/INFOHASH/ TORRENT_FILE/INFOHASH_HEX/ BASE32/DN/TR	parsing	P2	FR-001/002/003
FND-10	04 Â§6; 05 Â§8	DEFAULT_PORTS FastAPI 7187/Go 7189/ qBt 8080; DEFAULT_URLS; QBITTORRENT_ENDPOINTS map	api	P1/P4	OQ-01
FND-11	04 Â§6	Timing: DEBOUNCE MUTATION 500/ AUTO_SCAN 1000; REQUEST_TIMEOUTS DEFAULT 15000/ AUTH 10000/ADD 30000/DISCOVERY 3000; RETRY max 3 base 1000 max 30000 jitter 0.3; RATE_LIMIT 10/1000ms	api/ detection	P2/P4	config drift (OQ-19)
FND-12	04 Â§6	STORAGE_KEYS all bobalink_ prefixed (CONFIG/AUTH_STATE/ CREDENTIALS/DETECTED/HISTORY/ QUEUE/HEALTH/KEY_MATERIAL); only CONFIG/QUEUE/HEALTH actually written	data- model	P1	unused keys (OQ)
FND-13	04 Â§6	ENCRYPTION AES-GCM 256, IV 12B, salt 16B, PBKDF2 SHA-256 100000 iters, key version 1; passphrase source UNDEFINED in foundation	security	P4	OQ-15 critical
FND-14	04 Â§6	BADGE_COLORS (HEALTHY #4CAF50/ DEGRADED #FF9800/ERROR #F44336/ SCANNING #2196F3/DETECTED #9C27B0/DEFAULT); ICON_SIZES 16/32/48/128	ui	P6	FR-015/FR-021
FND-15	04 Â§4b	TS types: MagnetInfo, TorrentFile, ParsedTorrent, DetectedTorrent, SendResult, PageScanResult, ServerConfig(28 fields), ExtensionConfig, DEFAULT_CONFIG, AuthMethod, AutoDiscoveryConfig	data- model	P1	type duplication (OQ)
FND-16	04 Â§5	shared libs: crypto (encrypt/decrypt/sha256/ generateSecurePassphrase), errors (BobaLinkError+8 subclasses), events	data- model	P1	foundation modules

ID	Source	Fact / Requirement	Category	Phase	Notes
		(TypedEventEmitter 13-event EventMap), logger, storage (NamespacedStorage), utils (debounce/throttle/TokenBucket/ retryWithBackoff/yieldToBrowser/ processInChunks)			
FND-17	05 Â§2.1	BobaAPIClient: login/logout/getVersion/ addTorrentFromMagnet/ addTorrentFromFile/getTorrents/ deleteTorrents/pauseTorrents/ resumeTorrents; requestWithRetry+rate- limit+timeout	api	P4	FR-007/008/024/025
FND-18	05 Â§2.2	AuthHandler 4 methods (none/cookie/ api_key/basic); createCredentialsFromConfig decrypt point; MAX_CONSECUTIVE_FAILURES 3	security/ api	P4	FR-012; Â§11.4.10
FND-19	05 Â§2.3	OfflineQueue: DEFAULT_MAX_SIZE 50, PROCESS_INTERVAL 60000, ITEM_SEND_DELAY 500; enqueue FIFO evict/dequeue/processQueue/ startAutoProcessing	queue	P4	FR-014; ref max 50 vs spec 1000 (OQ-19); auth-gap (OQ)
FND-20	05 Â§2.4	HealthChecker: thresholds healthy<2000/ degraded<5000, MAX_FAILURES 2; checkServer/checkAllServers/ testConnection/autoDiscover(ports 8080/7187/7189)	api	P4	FR-024; FR-011
FND-21	05 Â§2.5	qBitTorrentAdapter: sendTorrent/ sendTorrents (250 ms delay)/ addTorrentFile/buildAddOptions/ mapContentLayout	api	P4	FR-007/008
FND-22	05 Â§3	bencode (zero-dep Uint8Array, sorted keys, sha1), magnet (parseMagnetUri xt/dn/tr/ws/ xl/xs/kt/as/mt + base32ToHex), torrent-file (parseTorrentFile infohash=sha1(encode(info)))	parsing	P2	FR-003/004/005; re- encode caveat
FND-23	05 Â§3 (note)	.torrent parser fully built but NEVER invoked (dead code) â€” sends upload raw File	parsing	P4	wire for dedup (OQ)
FND-24	05 Â§4	Scanner: BaseScanner (shadow-DOM querySelectorAllDeep, maxElements 10000), LinkScanner (site- specific+generic), TextScanner (TreeWalker), ScannerOrchestrator (MutationObserver debounce 500, per-site override, yieldToBrowser)	detection	P2/P3	FR-001/002; INS-03
FND-25	05 Â§4 (note)	NO chrome.tabGroups usage in ref; only intra-tab yieldToBrowser + per-tab tabTorrents Map â€” tab-group batching is net-new	tab- groups	P5	OQ-17; FR-009
FND-26	05 Â§4.5	Two overlapping selector tables: constants.SITE_SELECTORS (~21 domains, used by LinkScanner) vs site-	detection	P2	OQ; INS-03

ID	Source	Fact / Requirement	Category	Phase	Notes
		db.SITES (15 richer, used only for debounce) â€” reconcile			
FND-27	05 Â§5	Message protocol: ExtensionMessage{type,payload,requestId}; background handles scan-result/get-detected/send-torrent/get-config/set-config/health-check/test-connection/auto-discover/authenticate/scan-page/open-dashboard/queue-status/queue-process	api	P3	declared-but-unrouted types (OQ)
FND-28	05 Â§6	Background SW: lifecycle onInstalled/onStartup, initialize, initializeApiClient, sendTorrents, context menus, commands, alarms (keepalive ~20 s / health 5 min), badge/notifications, storage listeners	cross-browser	P3	DIM03-R2
FND-29	04 Â§2; 05 Â§6	commands: send-to-boba Ctrl+Shift+B, scan-page Ctrl+Shift+S, open-dashboard Ctrl+Shift+D (ref ships 3, differs from FR-017 B/L/G)	ui	P3	OQ-08
FND-30	05 Â§9; 05 banner	Â§11.4.10 BLUFF: options encrypts with FIXED passphrase â€œbobalink-extensionâ€¢; background decrypts with EMPTY â€œâ€¢â€” broken-by-design, MUST replace	security	P4	OQ-15 critical
FND-31	05 Â§2.5	fetch uses credentials:include/same-origin (qBt SID needs it) â€” conflicts DIM11-R8 credentials:omit â€” reconcile	security/ api	P4	DIM11-R8
FND-32	05 Â§6	Context menus bobalink-send (magnet/.torrent link patterns) / bobalink-scan / bobalink-dashboard	ui	P3	FR-016
FND-33	05 Â§7.1	Popup controls: status dot, Select All/Deselect/Refresh, torrent list (checkbox/name/type/infohash16/Sent), empty-state Scan Page, Send to Boba, progress overlay	ui	P3	FR-023; DIM10-R1
FND-34	05 Â§7.2	Options sections Servers/General/Advanced/About; server modal (name/url/auth/category/savepath/Test Connection); Auto Discovery card; General+Advanced toggles; Reset All	ui	P3	FR-023
FND-35	04 Â§4a	SQL dual-storage model: sql.js-over-storage.local (9 tables) vs STORAGE_KEYS JSON-blob + NamespacedStorage â€” pick one source of truth	data-model	P1	OQ dual-storage

10. Test-Type Coverage Requirements (toward 100% per Â§11.4.27)

ID	Source	Requirement	Category	Phase	Notes
TST-01			testing	P7	Â§11.4.27(A)

ID	Source	Requirement	Category	Phase	Notes
	06 Â\$types 1	Integration: real parserâ†’queueâ†’api-clientâ†’live qBt/proxy (7186/7187); no mocks outside unit			
TST-02	06 Â\$types 2	E2E functional: load real built MV3 bundle, resolve real ext id, drive popup/options/content with user-observable assertions	testing	P7	fixture builds dist + captures id (OQ)
TST-03	06 Â\$types 3	Automation/autonomous QA: self-driving NÃ—rerunnable (Â§11.4.98/Â§11.4.50); HelixQA banks	testing	P7	Â§11.4.116 sync channel
TST-04	06 Â\$types 4	Security/penetration: credential-leak audit (Â§11.4.10), XSS via magnet/dn, CSP, perm-scope, host-perm abuse, malformed injection, SID handling, crypto-at-rest	security/testing	P7	SEC-xx
TST-05	06 Â\$types 5	DDoS/load: 1000s magnet links/page, rapid scan cycles, queue enqueue burst, API flood/rate-limit under load	testing	P7	Â§11.4.85
TST-06	06 Â\$types 6	Scaling: large .torrent files, 1000s-item queue+persistency, large detected-set render, many servers	testing	P7	NFR-SCAL
TST-07	06 Â\$types 7	Chaos: storage failure/corruption mid-write, network drop/timeout/reorder, SW termination mid-op, corrupt-state recovery (Â§11.4.85 closed-set)	testing	P7	error-injecting mock needed
TST-08	06 Â\$types 8	Stress: sustained Nâ‰¥100/â‰¥30 s, concurrent Nâ‰¥10 enqueues/scans, boundary inputs, p50/p95/p99 latency	testing	P7	Â§11.4.85
TST-09	06 Â\$types 9	Performance: parse-time benchmarks, scan latency heavy DOM, popup render â€” captured evidence	performance/testing	P7	NFR-001..006
TST-10	06 Â\$types 10	Benchmark: formal suite + baselines + regression detection	testing	P7	
TST-11	06 Â\$types 11	UI tests: rendering correctness, badge counts, list render, modal state changes	testing	P7	beyond DOM-presence
TST-12	06 Â\$types 12	UX tests: detectâ†’selectâ†’send happy path, error feedback, empty/loading/error states, ally, keyboard nav, shortcuts	testing	P7	FR-020
TST-13			testing	P7	Â§11.4.27(B)

ID	Source	Requirement	Category	Phase	Notes
	06 Â\$types 13	Challenges (. / challenges / scripts /): real built ext vs real qBt+proxy, user-observable outcomes			
TST-14	06 Â\$types 14	HelixQA banks + autonomous QA sessions with captured evidence	testing	P7	Â\$11.4.27
TST-15	06 Â\$types 15; Â\$critical	Auth/auto-proc blocks, crypto-copy-not-module, non-functional e2e (hardcoded test-id), dead-asserts; every test fails vs no-op stub; bring UI/entrypoints INTO coverage	testing	P7	Â\$11.4/ Â\$11.4.1; reference unit+e2e is starting floor
TST-16	06 fixtures	Wire invalidMagnets (7 negative cases) + magnets.json/ base32 fixtures; reference test inventory (bencode/magnet/ torrent-file/api-client/queue/ scanner strong; crypto thorough-but-copy)	testing	P7	EDGE-23

11. Build / Release / Distribution

ID	Source	Requirement	Category	Phase	Notes
BLD-01	01 Â\$M.1	Build: npm install; npm run dev (WXT HMR); npm run build â†’ chrome/firefox/opera/edge MV3 prod; per-browser wxt build -b	build	P8	DIM12-R1
BLD-02	01 Â\$M.2	Test commands: jest â€œcoverage (80%), playwright; coverage thresholds; per-file targets	testing	P7	TST-15
BLD-03	01 Â\$M.3	Release: mainâ†’release branchâ†’bump manifestâ†’CHANGELOGâ†’full suiteâ†’git tagâ†’build allâ†’GitHub Releaseâ†’submit stores	build	P8	Â\$K.1 strip auto-submit (CI/CD)
BLD-04	01 Â\$M.4; 02 Dim12	Distribution: Chrome Web Store CRX (\$5), Firefox AMO XPI, Opera CRX, Edge, GitHub ZIP; per-store assets/2FA/review SLA	build	P8	DIM12-R4; DIM04-R2
BLD-05	01 Â\$M.5	Git: GitHub Flow; Conventional Commits (feat/fix/â€¦/scopes); branch prefixes â€œ” Boba: SSH remotes not HTTPS	build	P1/P8	FND-01 repo URL HTTPS
BLD-06	03 Â\$7	Pipeline stages Codeâ†’Lintâ†’Test(â‰¥80%)â†’Buildâ†’Packageâ†’Deploy;	build	P7/P8	diagram #11

ID	Source	Requirement	Category	Phase	Notes
		quality gates 0 ESLint errors/valid manifest/<5MB			
BLD-07	02 Dim12; 04 Â§7	release-please + Husky pre-commit + lint-staged + web-ext sign “ Boba: DROP Husky (git hooks forbidden)	build	P8	Â§K.1
BLD-08	01 Â§K.1; 03 Â§9; 04 Â§7; 06 Â§10	CI/CD REMOVAL (Boba Hard-Stop): strip ALL .github/workflows/*.yml (ci.yml, release.yml), Husky hooks, “œenforced in CI“ , auto store-submission “’ convert to manual ./ci.sh-style script Manual version-bump touches package.json + wxt.config manifest + both SQL files + app_metadata/extension_config schema_version rows together	build	P8	MANDATORY constraint
BLD-09	04 Â§7		build	P8	release.yml sed

12. Open Questions / Ambiguities (consolidated, for conductor)

ID	Source	Question	Category	Phase	Notes
OQ-01	01 Â§T1; 03 Â§9.4; 04 Â§8.1; 05 Â§9.2	Real Boba port mapping: 8443/8080“’7186/7187/7189; qBt direct“’proxy 7186? Boba REST/SSE“’7187? search“’7189? Re-point all discovery defaults	api	P1/P4	HIGH/critical
OQ-02	01 Â§T2; 03 Â§9.1; 04 Â§7; 06 Â§10	CI/CD removal shape “” manual ./ci.sh	build	P8	HIGH; BLD-08
OQ-03	01 Â§T3; 06 Â§1	Jest vs Vitest (Boba frontend uses Vitest; crypto.test.ts already Vitest)	testing	P7	MED
OQ-04	01 Â§T4; 01 Â§O.3	Two divergent source layouts (src/parser/scanner/api/shared vs src/modules/utis/content-scripts) “” pick canonical	data-model	P1	MED
OQ-05	01 Â§T5; 01 Â§P.1	Two divergent phase plans (plan.md 10 vs impl-plan 8) “” use 8-phase	build	P1	MED; this matrix uses 8
OQ-06	01 Â§T6; 01 Â§D.7	Data-model vocabulary drift: source_type adds torrent-url; status completed vs dead-letter; auth_method adds none drops custom-header “” reconcile enums	data-model	P1	MED
OQ-07	01 Â§T7	Magnet param superset: FR-003 xt/dn/tr/xl/ws/x.pe vs impl-plan	parsing	P2	

ID	Source	Question	Category	Phase	Notes
		xs/as/kt/mt (no x.pe) â€” which implemented			LOW; ref impl xt/dn/tr/ws/xl/ xs/kt/as/mt
OQ-08	01 Â§T8; 04 Â§2; 05 Â§6	Keyboard-shortcut count 3 vs 4; ref ships B/S/D, FR-017 says B/L/ ui G; Ctrl+Shift+L collision		P3	LOW; FND-29
OQ-09	01 Â§T9; 02 INS-07; 04 Â§8	host_permissions scope: https://*/ (ref) vs activeTab minimal (NFR-012) â€” store-review/ privacy	security	P1	LOW-MED
OQ-10	01 Â§T10; 04 Â§1	LICENSE: impl-plan Apache 2.0 vs package.json MIT â€” confirm	build	P1	LOW
OQ-11	01 Â§T11	Naming: â€œBobaLinkâ€” reference name â€” confirm Boba repo extension name + icon	build	P1	LOW
OQ-12	01 Â§T12; 02 Â§509-510; 03 Â§9.6	Search/SSE scope: does Boba 7187 expose /api/v1/ torrents/search+SSE as spec, or call existing merge routes? Confirm vs live / openapi.json	api	P4	MED; CV-04
OQ-13	01 Â§T13	Boba /api/v1/health vs real /health â€” confirm path/port	api	P4	LOW-MED
OQ-14	04 Â§8.3; 05 Â§9.3	Dual storage model (sql.js 9-table vs JSON-blob NamespacedStorage) â€” one source of truth	data- model	P1	FND-35
OQ-15	04 Â§8.2; 05 Â§9.1	Encryption passphrase provenance UNDEFINED; ref fixed/empty-passphrase BLUFF â€” design real key source (master pw prompt / OS keychain / delegate to boba-jackett /config/ boba.db)	security	P4	CRITICAL Â§11.4.10; FND-30
OQ-16	05 Â§9.9	Time-salted detection ids (Date.now() in hashString) â†’ unreliable cross-scan dedup â†’ use infohash-derived stable ids	parsing	P2	FR-006
OQ-17	05 Â§9.8; 04 Â§4.5	Tab-group batching does NOT exist in ref â€” net-new design	tab-groups	P5	FND-25; FR-009
OQ-18	05 Â§9.2	Boba integration only port-deep â€” client speaks only qBt WebUI v2; no merge-search/proxy(7186)/ boba-jackett(7189) code; define how ext reaches Bobaâ€™s own endpoints	api	P4	DIM01-R2
OQ-19	01 Â§T6; 05 Â§2.3; 04 Â§8.5	Config-value drift: queue max 50(ref)/1000(spec)/100(Dim09); retries 3/5; RATE_LIMIT 10 vs DB seed 30; autoScanDelay 2000	queue/api	P4	FR-014/025

ID	Source	Question	Category	Phase	Notes
OQ-20	05	vs DEBOUNCE 1000 " pick canonical Offline queue processQueue never authenticates " 403 loop against authed server " wire auth	queue	P4	FND-19
OQ-21	05	Two get-detected response shapes; unrouted message types (send-result/get-auth-state/health-result/show-notification/update-badge/torrent-detected/selection-change) " unify/decide	api	P3	FND-27
OQ-22	05	set-cookie SID parsing brittle across CORS; CORS/host-perm story for cross-origin localhost; manifest/commands/icon rasterization to author	api/build	P3/P4	DIM07-R4
OQ-23	06	E2E extension-loading fixture (build dist " persistent context "load-extension " capture id); Chromium-only vs Firefox MV3; global-setup/teardown contract; live backends for integration	testing	P7	TST-02/03
OQ-24	03	Single-send routing dual-write FastAPI+Go ambiguity; batch endpoint path (/ api / torrents / batch vs direct qBt); extension_config " i, Ždiscovered FK conceptual; X-Api-Key+Basic together vs per-service	api/data-model	P4	diagram inconsistencies

COVERAGE LEDGER

Item counts by category (every row classified)

Category	Count
detection	22
parsing	27
api	41
queue	12
ui	24
security	35
privacy	4
i18n	3
build	18
data-model	22

Category	Count
cross-browser	11
tab-groups	12
performance	12
reliability	3
scalability	4
maintainability	1
testing	23
(rows are multi-tagged; primary-category tallies above)	

Item counts by section (authoritative row totals)

Section	Rows
1. Functional Requirements	25 (FR-001..FR-025)
2. Non-Functional Requirements	15 (NFR-001..NFR-015) + 4 scalability/maint (NFR-SCAL-1..3, NFR-MAINT-1)
3. Research-Dimension Decisions	56 (DIM01-R1..DIM12-R4 across all 12 dims)
4. ER Entities / Data Model	8 (6 ER entities + ER-REL + ER-SQL-extra)
5. Edge Cases	24 (EDGE-01..EDGE-24)
6. Security / STRIDE	11 (SEC-01..SEC-11)
7. Insights	10 (INS-01..INS-10)
8. Cross-Verification	5 (CV-01..CV-05)
9. Foundation Facts	35 (FND-01..FND-35)
10. Test-Type Coverage	16 (TST-01..TST-16)
11. Build / Release	9 (BLD-01..BLD-09)
12. Open Questions	24 (OQ-01..OQ-24)
TOTAL traceable items	245

v1 vs v2 split

- **v1 (in-scope for initial release):** 240 items – all FRs, all NFRs, all dimension decisions except deferred, all ER entities except search_cache, all edge cases, all security, 8 of 10 insights, all foundation facts, all test types, all build (CI/CD converted to manual), all open questions resolved during build.
- **v2 (explicitly deferred):** 5 items.

Items intentionally deferred to v2 (with rationale)

ID	Item	Rationale
INS-08	Multi-Client Gateway (TorrentClient interface: Transmission/Deluge/rTorrent adapters)	MEDIUM confidence; adds complexity; v1 targets Boba/qBittorrent only. v1 architecture leaves a <code>TorrentClient</code> seam but ships ONE adapter. (file 02 Insight 8; file 01 – 509 OQ-7.)
ER-search_cache	Search-result cache entity (TTL™d, standalone, no FK)	Search aggregation is gated to Boba 1.2+ (file 01 – R

ID	Item	Rationale
		feature-availability); not needed for the core detectâ†’send v1 flow. Storage backend unresolved (OQ-12/OQ-24).
DIM01-R2 (search subset)	Boba merge-search /api/v1/search + SSE /api/v1/search/stream/{id} live-progress in popup	â€œSearch aggregation = Boba 1.2+ onlyâ€¢; SSE progress = Boba 1.1+ only (file 01 Â§R). v1 ships detectâ†’send + qBt-direct + basic health; rich search/SSE is a post-v1 enhancement. (Core /api/v1/download + /api/v1/magnet stay v1.)
DIM10-R5	Side panel (Chrome 114+, open 116+, close 141+; cannot auto-open)	LOW confidence (CV-05); browser-floor too high vs Chrome 88 target; popup covers v1 UX. Optional surface.
DIM02-R3 / CV-03	qBittorrent API-key (Bearer qb_t_) auth, v5.2.0+ only	LOW confidence; only on newest qBt; cookie auth is the safe v1 baseline. API-key path deferred until version floor justifies it.

NOTE: deferral means â€œnot built in v1,â€¢ NOT â€œdropped.â€¢ Each carries a v1 seam (TorrentClient interface, search-cache storage decision, SSE client stub, side-panel optional entry, API-key auth branch) so v2 is additive.

Cross-check: completeness of the mandated baselines

- **25 FR â€œ ALL represented:** FR-001â€¢FR-025 each have exactly one row (Â§1). â€œ
- **15 NFR â€œ ALL represented:** NFR-001â€¢NFR-015 each have one row; scalability/maintainability sub-clauses captured as NFR-SCAL-1..3 + NFR-MAINT-1 (Â§2). â€œ
- **12 Dimensions â€œ ALL represented:** DIM01 (4), DIM02 (4), DIM03 (5), DIM04 (4), DIM05 (3), DIM06 (4), DIM07 (5), DIM08 (6), DIM09 (9), DIM10 (6), DIM11 (10), DIM12 (4) â€œ every dimension has â‰¥3 consolidated decision rows; 56 total (Â§3). â€œ
- **6 ER entities â€œ ALL represented:** ER-extension_config, ER-discovered_torrents, ER-download_queue, ER-server_config, ER-send_history, ER-search_cache; + ER-REL relationships + ER-SQL-extra (app_metadata/site_selectors/queue_log/statistics) superset (Â§4). â€œ
- **Edge cases:** all 16 file-01-Â§I cases (EDGE-01..21 incl.Â multi-item lines) + Dim06/07/05/08 cases (EDGE-22..24) (Â§5). â€œ
- **Insights:** all 10 (INS-01..10) (Â§7). â€œ
- **Cross-verification:** 4 conflicts + low-confidence set (CV-01..05) (Â§8). â€œ

Items that could NOT be placed (OPEN)

NONE. Every FR, NFR, dimension-decision, ER entity, edge case, insight, cross-verification conflict, foundation fact, test-type, build step, and open question from all six analysis files has been assigned a stable ID and a row. No source item is unaccounted for.

The reference-deliverable internal inconsistencies (dual source layouts, dual storage models, two phase plans, port mismatches, config-value drift, the Â§11.4.10 fixed/empty-passphrase bluff, dead/unrouted code, missing E2E fixtures, CI/CD-vs-Boba-Hard-Stop) are NOT untraceable items â€” they are captured as explicit OQ-xx / FND-xx rows requiring a conductor decision during the corresponding phase, per Â§11.4.118 enumerated coverage and Â§11.4.66 interactive clarification.

End of C-traceability-matrix.md â€” anti-skip backbone for the implementation plan.